

Humberto Sandmann



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Computer scientist and educator with a D.Sc. from the University of São Paulo and a research stay at the Max Planck Institute for Dynamics and Self-Organization. My research bridges bio-inspired neural computing, nonlinear dynamics, and pattern recognition, investigating how biological neuronal mechanisms can inform computational intelligence. With more than a decade of university teaching alongside senior industry practice in software architecture, machine learning, and high-performance computing, I bring research depth and applied engineering experience into the classroom.

Research Interests

DeFi, Smart contracts and decentralized applications

Small-world networks and complex systems

Computation in large-scale systems

Artificial neural networks and deep learning

Nonlinear systems and network dynamics

Synchronization of coupled oscillators

Education

[mar.2012](#)
[↑](#)
[mar.2007](#)

D.Sc.
 thesis
[networks](#)

Polytechnic School of the University of Sao Paulo
 Spike patterns and computation in dynamical neural



doi [10.11606/T.3.2012.tde-05092012-165022](#)
 advisor [Prof. Dr. Emilio Del Moral Hernandez](#)
 presentation [slides](#)
 supported by [CNPq - grant 140632/2007-7](#)
 (mar.2007 → dec.2009) $\wedge$ (sep.2010 → feb.2011)

[sep.2010](#)
[↑](#)
[aug.2009](#)

D.Sc. Internship
[Selbstorganisation](#)
 advisor [Prof. Dr. Marc Timme](#)
 supported by [CAPES - grant BEX 4076-09-5](#)

Max Planck Gesellschaft für Dynamik und



[jan.2010](#) → [aug.2010](#)
[Pró-Reitoria da Pós-Graduação da USP](#)
[aug.2009](#) $\wedge$ [dec.2009](#) (flight tickets GRU $\nrightarrow$ FRA)
[José Humberto Sandmann - my father](#)
[aug.2009](#) → [dec.2009](#) (German language classes/exams)
[Max Planck Institute - Networks Dynamics Group](#)
[aug.2009](#) → [nov.2009](#) (accommodation in Goettingen)

[jun.2006](#)
[↑](#)
[feb.2003](#)

M.Sc.
 dissertation
[neuro-fuzzy systems](#)

Polytechnic School of the University of Sao Paulo
 Prediction of time series using architecture based on



doi [10.11606/D.3.2006.tde-01042009-095125](#)
 advisor [Prof. Dr. Marco Túlio Carvalho de Andrade](#)

[dec.2002](#)
[↑](#)
[feb.1999](#)

B.Sc.
 Faculdade de Engenharia Industrial
 Computer Science Bachelor

[dec.1998](#)
[↑](#)
[feb.1996](#)

Escola Técnica Jorge Street
 Technician in Industrial Computation

Academic Appointments & Teaching

now

↑

jun.2018

Inspira

Sao Paulo, Brazil

now ↑ feb.2020	Associate Professor		
	Platform, Microservices and APIs	Spring Cloud, Spring Boot, PostgreSQL, Redis, RabbitMQ, Kafka, Docker, K8s	2026.1, 2025.2, 2025.1, 2024.1
	Coding Interviews	Course to prepare for coding interviews.	2026.1
	Machine Learning	PCA, t-SNE, UMAP, KNN, K-Means, Decision Trees, Random Forests, Support Vector Machines, Classification, Regression, Clustering, EDA, Pre-processing, Feature Engineering	2026.1
	Neural Networks and Deep Learning	Artificial Neural Networks, Perceptron, Multi Layer Perceptron, Convolutional Neural Networks, GAN, VAE, CLIP, Stable Diffusion	2025.2
	TecWeb	JS, React	2025.2
	Cloud Computing	MAAS, Juju, OpenStack, K8s, IaaS, PaaS , Computer Networks	2025.1, 2024.2
	Design of Software	Python	2025.0, 2024.1, 2023.2, 2022.1, 2021.2, 2021.1
	Information System for Administration and Economy	Excel, VBA, Python, NumPy, Pandas, Stochastic Processes, RoI	2023.2, 2023.1, 2022.2

	Spreadsheet Automation in Excel	Excel, VBA	2021.1, 2020.1
jan.2020	Assistant Professor		
↑			
aug.2018	Design of Software	Python	2019.2, 2019.1
	Foundations for Physical Communication	Python, Fourier Transform, Wavelet Transform, Signal Processing	2018.2

dec.2025

↑

jun.2014

[ESPM](#)

Sao Paulo, Brazil

Associate Professor

[Machine Learning](#)

Machine Learning, PCA, KNN, K-Means, Decision Trees, Random Forests, Support Vector Machines, Naïve Bayes, Page Rank, PySpark, Classification, Regression, Clustering, Pre-processing, Feature Engineering

2025.2

[Applied Mathematics](#)

Propositional Logic, Boolean Algebra, Set Theory, Polynomials, Functions, Sequences, Series, Logarithm, Combinatorial Analysis, Limits, Derivatives, Linear Algebra, Matrix Algebra, Graph Theory

[2025.1](#), [2024.1](#),
[2023.2](#), [2023.1](#),
[2022.2](#), [2022.1](#),
[2021.2](#), [2021.1](#),
2020.2, 2020.1

[Oriented Object Programming](#)

Java, Design Patterns, Docker, Spring Cloud, Microservices

2025.1, 2024.1,
2023.1, 2022.1,
2021.1

Statistics

Statistics, Probability, Random Variables, Distributions, Sampling, Estimation, Hypothesis Testing, Regression Analysis

[2023.2](#)

Software Quality

Software Quality, Software Testing, Software Architecture, Software Engineering, Software Development Life Cycle, Agile Methodologies, Test-Driven Development, Behavior-Driven Development, Test Automation, Continuous Integration, Continuous Delivery

2024.2, [2022.2](#)

Architecture and Operating Systems

Computer Architecture, Operating Systems, Binary and Hexadecimal Number Systems, Number Representation, Data Representation, Assembly Language, Memory Management, Process Management, File Systems

[ESPM 16-bits](#)
2021.1, 2020.2,
2020.1, 2019.2,
2019.1, 2018.2,
2018.1, 2017.1,
2016.2, 2016.1,
2015.2, 2015.1,
2014.2

Artificial Intelligence and Society	Artificial Intelligence, Machine Learning, Deep Learning, Neural Networks, Computer Vision, Natural Language Processing, PCA, KNN, K-Means, GANs, Transformers, LSTMs, BERT, Ethics, Bias	2022.1 (IR), 2021.1 , 2020.2 , 2020.1 , 2019.2
Introduction for Artificial Intelligence	Machine Learning, Deep Learning, Neural Networks, Computer Vision, Natural Language Processing, PCA, KNN, K-Means, GANs	2020.2 , 2020.1 , 2019.2 , 2019.1

apr.2015 [Toledo Prudente Centro Universitário](#)

↑

Presidente Prudente, Brazil

mar.2015 Visiting Professor

MBA: Big Data and Machine Learning	Big Data, Hadoop, Hive, HBase, MongoDB, Neo4j, Solr, R, Octave, Self-Organizing Maps, Bayesian Decision Trees, Multilayer Perceptron	2015.1
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dec.2015

[FIAP](#)

↑

Sao Paulo, Brazil

mar.2012

Associate Professor

Distributed systems	Sockets, RMI, SOAP, Microservices, J2EE, Design Patterns	2015
MBA: Big Data and No-SQL Modelling	Big Data, Hadoop, Hive, HBase, MongoDB, Neo4j, Solr	2014.2
Data Structure and Algorithm II	C, Data Structure, Algorithm, Complexity, Linked List, Stack, Queue, Tree, Red-Black Tree, AVL Tree, B-Tree, Hash Table, Graph, Dijkstra Algorithm, Prim Algorithm	2013
Algorithms	C, Programming, Algorithm, Complexity	2013, 2012.2
Database I and II	SQL, PL/SQL, Oracle, Transaction, Concurrency Control, Locking, Deadlock, Recovery, Normalization, ER Model, Relational Model, Object-Relational Model, Trigger, View, Index, Stored Procedure, Function, Package, Cursor, Sequence, Synonym	2013, 2012
Programming III	Pointers, C, C++, Object-Oriented Programming, Inheritance, Polymorphism, Encapsulation, Abstract Class, Interface, Exception Handling	2012

jul.2013	Universidade Cruzeiro do Sul		
↑	Sao Paulo, Brazil		
mar.2012	Associate Professor		
	Mobile Programming	Android	2012.2
	Computer Architecture	Computer Architecture, Operating Systems, Binary and Hexadecimal Number Systems, Number Representation, Data Representation, Assembly Language, Memory Management, Process Management, File Systems	2012.1
	Software Engineering	Software Engineering, Software Development Life Cycle, Agile Methodologies, Test-Driven Development, Behavior-Driven Development, Test Automation	2012.1
	Introduction to Programming	C, Programming, Algorithm	2012.1
	Database II	SQL, PL/SQL, Oracle, Transaction, Concurrency Control, Locking, Deadlock, Recovery, Normalization, ER Model, Relational Model, Trigger, View, Index, Stored Procedure, Function, Package, Cursor, Sequence	2013.1, 2012.1
jul.2011	Universidade Nove de Julho		
↑	Sao Paulo, Brazil		
mar.2011	Temporary Professor		
	Computer Architecture	Computer Architecture, Operating Systems, Binary and Hexadecimal Number Systems, Number Representation, Data Representation, Assembly Language, Memory Management, Process Management, File Systems	2011.1
	Introduction to Programming	C, Programming, Algorithm	2011.1
	Web Programming	HTML, CSS, JavaScript	2011.1

jul.2011

↑

mar.2008

[Polytechnic School of the University of Sao Paulo](#)

Sao Paulo, Brazil

Teaching Assistant

Experimental Pattern
Recognition, Modeling
and Neurocomputing

Logic Gates, Linear Regression, Pattern
Classification, [UC Irvine Machine Learning
Repository](#), [NIST Data Gateway](#), Multilayer
Perceptron, EEG Signals, PCA

[2011.1*](#)

Modeling for Signal
Processing: Neural
Networks and Learning

Artificial Neural Networks, Multilayer
Perceptron, Backpropagation, Forecasting,
Time Series, Classification

[2009.1,](#)
[2008.1*](#)

Laboratory of the
Foundations of
Computing
Engineering

Computer Architecture, Assembly
Language, Memory Management, Process
Management, Register, Stack, Turing
Machine, Finite State Machines, Von
Neumann Architecture, Java, Orientated
Object Programming

[2008.1†](#)

* grants from CAPES (PAE)

† grants from FAPESP

Professional Experience

mar.2025 ↑ feb.2006	H-Sandmann Sao Paulo Area, Brazil Founder	Machine Learning, Software Architecture.
oct.2023 ↑ mar.2012	SelsanTech Sao Paulo Area, Brazil Co-Founder	Machine Learning, Software Architecture.
jun.2019 ↑ may.2018	RockSpoon Sao Paulo Area, Brazil - Palo Alto, United States Senior Software Architect and Machine Learning Engineer	POS, ERP, Payroll, JavaEE, Postgres, Software Architecture, OpenCV, Machine Learning, Computer Vision, GoLang, Android, Kotlin.
jul.2016 ↑ oct.2015	Banco Votorantim Sao Paulo Area, Brazil Contractor: Senior Consultant for Computational Optimization	RoE, Banking Business, High-Performance Computing, J2EE, Spring Boot, Automating Testing, Jenkins.
nov.2011 ↑ aug.2011	TecBan Sao Paulo Area, Brazil Contractor: Senior Consultant for Computational Optimization	ATM, J2EE, InMemory Database, High-Performance Computing.
jun.2007 ↑ feb.2006	ABN Amro Bank Sao Paulo, Brazil Contractor: Senior Software Engineer/Architect	Stock Market, Trading, Core Banking, Software Architecture, Cryptography, HSM, BouncyCastle, PKI, PCKS, Java, Servlet, Database.
jan.2006 ↑ aug.2005	Politec Sao Paulo, Brazil Senior System Analyst	J2EE, Software Quality, Software Architecture, Banking Business.
jul.2005 ↑ jul.2004	EverSystems Sao Paulo, Brazil Senior Software Engineer/Architect - BankBoston - CitiBank - Elly Lilly	PKI, BouncyCastle, Cryptography, HSM, Core Banking, Stock Market, Java, Servlet, Database, Communication Protocol, Synchronization and Replication.

mar.2003	Central Online	Object-Oriented Programming,
↑	Sao Paulo, Brazil	LMS .
sep.2002	Senior System Analyst	
sep.2002	Datamace	Payroll, ERP, Software
↑	Sao Paulo Area, Brazil	Architecture.
oct.1999	Senior System Analyst	
oct.1999	Heraeus Electro-Nite	Compliance, ERP, PCP.
↑	Sao Paulo Area, Brazil	
mar.1998	Computer Programmer	

Publications

Selected peer-reviewed publications. Full and up-to-date list: [ORCID 0000-0001-7758-695X](https://orcid.org/0000-0001-7758-695X).

E. Del-Moral-Hernandez; H. Sandmann; G. Araujo; **Context dependent pattern recognition - A framework for hybrid architectures bridging chaotic neural networks based on Recursive Processing Elements and symbolic information**. IEEE Transactions on Neural Networks, 2009. doi: [10.1109/IJCNN.2009.5179061](https://doi.org/10.1109/IJCNN.2009.5179061).

L.A. da Silva; H. Sandmann; E. Del-Moral-Hernandez; **A self-organizing architecture of recursive elements for continuous learning**. IEEE Transactions on Neural Networks, 2008. doi: [10.1109/IJCNN.2008.4634190](https://doi.org/10.1109/IJCNN.2008.4634190).

E. Del-Moral-Hernandez; H. Sandmann; L.A. da Silva; **Pattern recovery in networks of recursive processing elements with continuous learning**. IEEE Transactions on Neural Networks, 2004. doi: [10.1109/IJCNN.2004.1379877](https://doi.org/10.1109/IJCNN.2004.1379877).

Grants & Funding

CNPq	Doctoral fellowship – grant 140632/2007-7	mar.2007 → dec.2009 ^Λ sep.2010 → feb.2011
CAPES	Doctoral internship abroad (Max Planck Institute) – grant BEX 4076-09-5	jan.2010 → aug.2010
CAPES (PAE)	Teaching assistantship fellowship – Polytechnic School, USP	2008, 2011
FAPESP	Teaching assistantship fellowship – Polytechnic School, USP	2008

Academic Service & Memberships



IEEE — Institute of Electrical and Electronics Engineers

Member #60216475

- **Computational Intelligence Society (CIS)** — member since 2004; Brazilian committee member (apr.2009 → aug.2009 ∧ oct.2010 → mar.2012)
- **Signal Processing Society (SPS)** — 2005 → 2009
- **Engineering in Medicine and Biology Society (EMBS)** — 2005 → 2009



INNS — International Neural Network Society

Student member #3951

Languages

Portuguese native

English advanced

German **B1**